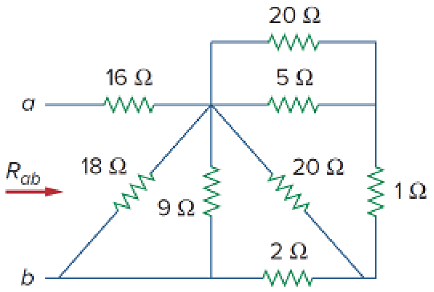
Chapter 2, Problem 10PP

Problem

Find *Rab* for the circuit in Fig. 2.39.

Figure 2.39:



Step-by-step solution

Step 1 of 8

The circuit is shown in Figure 1.

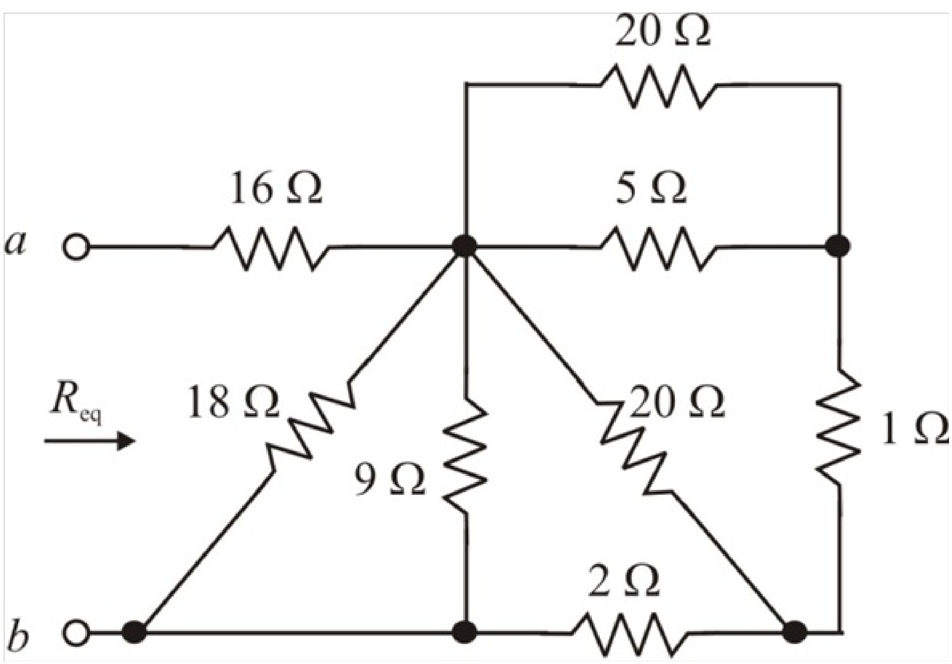


Figure 1

Step 2 of 8

To get R_{ab}, we combine resistors in series and in parallel.

The 20? and 5? resistors are in parallel. Their equivalent resistance is:

$$\begin{aligned} 20\Omega \parallel 5\Omega &= \frac{(20)(5)}{20+5} \\ &= 4\Omega \end{aligned}$$

Step 3 of 8

The circuit in Figure 1 is reduced to that in Figure 2.

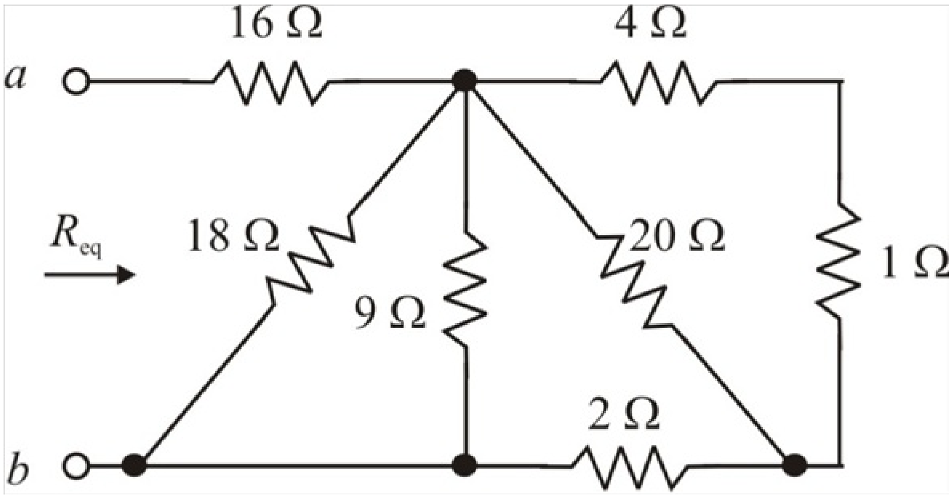


Figure 2

Step 4 of 8

The 4? and 1? resistors are in series. Their equivalent resistance is:

$$4\ \Omega + 1\ \Omega = 5\ \Omega$$

The circuit in Figure 2 is reduced to that in Figure 3.

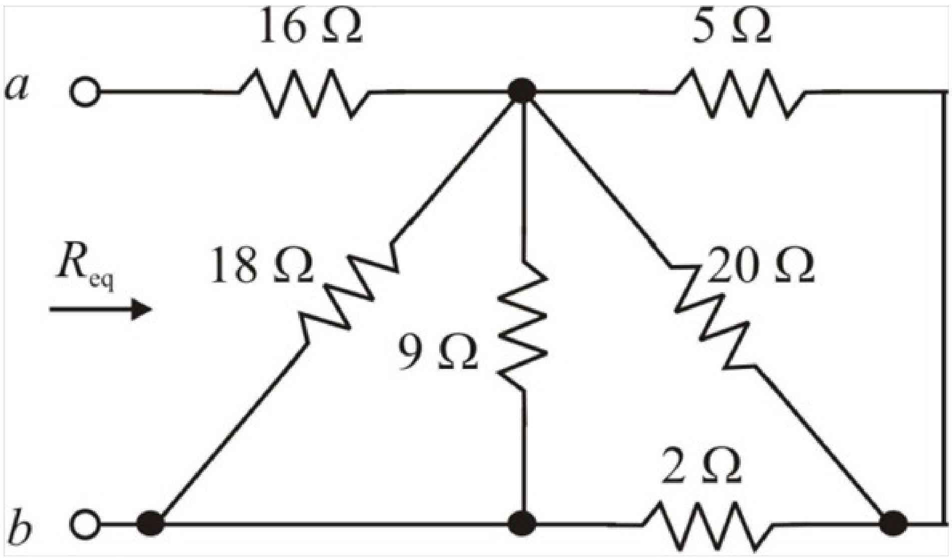


Figure 3

Step 5 of 8

The 20? and 5? resistors are in parallel. Their equivalent resistance is:

$$\begin{aligned} 20\Omega \,\|\, 5\Omega &= \frac{(20)(5)}{20+5} \\ &= 4\ \Omega \end{aligned}$$

Step 6 of 8

The circuit in Figure 3 is reduced to that in Figure 4.

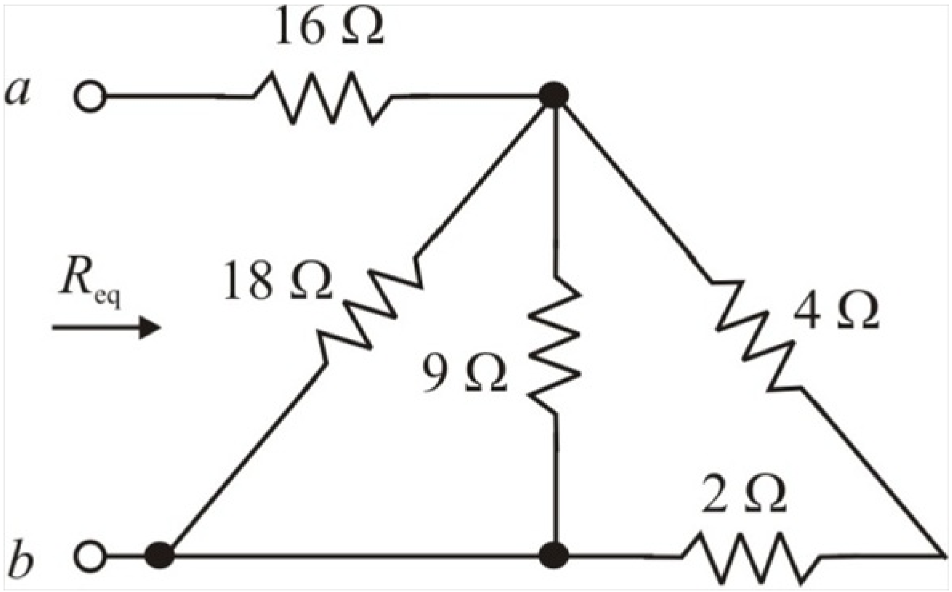


Figure 4

Step 7 of 8

The 4? and 2? resistors are in series. Their equivalent resistance is:

$$4\Omega + 2\Omega = 6\Omega$$

The circuit in Figure 4 is reduced to that in Figure 5.

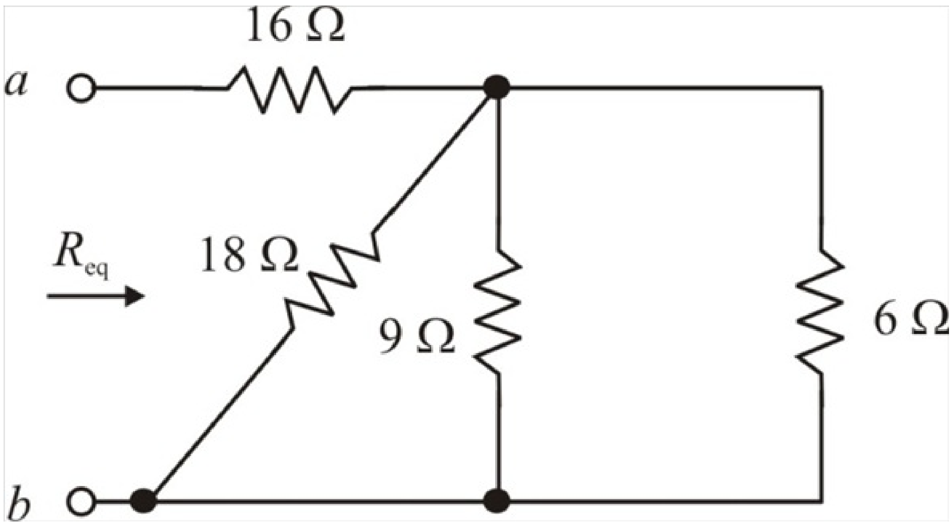


Figure 5

Step 8 of 8

The 6? and 9? resistors are in parallel. Their equivalent resistance is:

$$6\Omega \,||\, 9\Omega = \frac{(6)(9)}{6+9}$$
$$= 3.6\Omega$$

This 3.6? and 18? resistors are in parallel. Their equivalent resistance is:

$$3.6\Omega \,||\, 18\Omega = \frac{(3.6)(18)}{3.6+18}$$
$$= 3\ \Omega$$

This 3? and 16? resistors are in series. Their equivalent resistance is:

$$3\Omega + 16\Omega = 19\Omega$$

Therefore, the equivalent resistance is $\boxed{19\ \Omega}$.